

A Counterexample to Conjecture 5.1 of Lai–Shi

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Source

The source is Chun-Kit Lai and Ruxi Shi, *When is the fractal uncertainty principle for discrete Cantor sets most uncertain?*, arXiv:2501.00864.

Conjecture 5.1 of the source states that if $(\mathcal{A}, \mathcal{B})$ is a distributed spectral pair in $\{0, 1, \dots, M-1\}$, then $(\mathcal{A}, \mathcal{B})$ is either a spectral pair in $\mathbb{Z}_M$ or a spectral pair in $\mathbb{Z}_{M^2}$.

Definitions Used

For a finite set $\mathcal{A} \subset \mathbb{Z}$ write

$$P_{\mathcal{A}}(x) = \sum_{a \in \mathcal{A}} x^a.$$

The source's distributed spectral condition says that for every nonzero difference $d = b_1 - b'_1$ with $b_1, b'_1 \in \mathcal{B}$, either

$$P_{\mathcal{A}}\left(e^{2\pi i d/M}\right) = 0$$

or, for every $e = b_2 - b'_2$ with $b_2, b'_2 \in \mathcal{B}$,

$$P_{\mathcal{A}}\left(e^{2\pi i (d+Me)/M^2}\right) = 0.$$

The sign convention in the Fourier transform is immaterial for these zero tests.

Proof Intuition

The source gives an example outside the digit range whose obstruction is carried by a second-level cyclotomic zero. The construction below keeps the same idea but folds it into the legal digit range. The set $\mathcal{A}$ is chosen so that $P_{\mathcal{A}}$ has zeros of orders 2, 4, 6, 12, and also the extra second-level order 36. The difference set of $\mathcal{B}$ is arranged so that all but two differences are already killed modulo 24, while the two remaining differences become order-36 roots after every distributed translate.

Theorem 1. *Let*

$$M = 24, \quad \mathcal{A} = \{0, 3, 18, 21\}, \quad \mathcal{B} = \{0, 4, 16, 20\}.$$

Then $(\mathcal{A}, \mathcal{B})$ is a distributed spectral pair in $\{0, 1, \dots, M-1\}$, but it is neither a spectral pair in $\mathbb{Z}_M$ nor a spectral pair in $\mathbb{Z}_{M^2}$.

Proof. We first compute

$$P_{\mathcal{A}}(x) = 1 + x^3 + x^{18} + x^{21} = (1 + x^3)(1 + x^{18}) = \Phi_2(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x)\Phi_{36}(x).$$

Consequently $P_{\mathcal{A}}(\zeta) = 0$ whenever ζ is a root of unity of order

$$2, 4, 6, 12, \text{ or } 36. \tag{1}$$

The difference set of $\mathcal{B}$ is

$$\mathcal{B} - \mathcal{B} = \{0, \pm 4, \pm 12, \pm 16, \pm 20\}.$$

For $d = \pm 4, \pm 20$, the root $e^{2\pi i d/24}$ has order 6, and for $d = \pm 12$ it has order 2. Hence the first alternative in the distributed spectral condition holds for these differences by (1).

It remains to check $d = \pm 16$. Write any $e \in \mathcal{B} - \mathcal{B}$ as $e = 4s$, where

$$s \in \{0, \pm 1, \pm 3, \pm 4, \pm 5\}.$$

Then

$$d + 24e = \pm 16 + 96s = 16(\pm 1 + 6s).$$

The integer $\pm 1 + 6s$ is relatively prime to 36, so

$$\gcd(24^2, d + 24e) = 16.$$

Therefore $e^{2\pi i(d+24e)/24^2}$ has order $24^2/16 = 36$, and the second alternative in the distributed spectral condition holds for every $e \in \mathcal{B} - \mathcal{B}$ by (1). This proves that $(\mathcal{A}, \mathcal{B})$ is distributed spectral.

It is not a spectral pair in $\mathbb{Z}_{24}$: taking $d = 16 \in \mathcal{B} - \mathcal{B}$, the root $e^{2\pi i 16/24}$ has order 3, which is not among the zero orders in (1), so $P_{\mathcal{A}}(e^{2\pi i 16/24}) \neq 0$.

It is not a spectral pair in $\mathbb{Z}_{24^2}$ either: taking $d = 4 \in \mathcal{B} - \mathcal{B}$, the root $e^{2\pi i 4/24^2}$ has order $24^2/\gcd(24^2, 4) = 144$, which is not among the zero orders in (1). Hence $P_{\mathcal{A}}(e^{2\pi i 4/24^2}) \neq 0$. $\square$

Verification Notes

The accompanying script `code/verify_counterexample.py` checks the same facts using exact integer arithmetic: it computes orders as $N/\gcd(N, t)$ and tests membership in the cyclotomic zero-order set $\{2, 4, 6, 12, 36\}$.

Novelty Check

Local searches in the run indexes, parsed arXiv sources, and metadata for 2501.00864, **distributed spectral pair**, the source title, and the explicit sets found no prior packet or occurrence of this example. Web searches for the explicit sets and phrases such as "**distributed spectral pair**" "**counterexample**" " **$\mathbb{Z}_{M^2}$** " found no prior answer. The counterexample was found by a bounded brute-force search: no counterexample appeared for $M \leq 15$, and this one appeared in the targeted $M = 24$, four-point search.

Reference

C.-K. Lai and R. Shi, *When is the fractal uncertainty principle for discrete Cantor sets most uncertain?*, arXiv:2501.00864, 2025.